ETAS RTA-BSW v12.7.0

RTA-Hwd (Fray) Reference Guide

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ETAS RTA-BSW 12.7.0 - RTA-Hwd (Fray) Reference Guide R01 EN – 05.2025

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1 FlexRay Transceiver Driver

Module name	FrTcrv
Package	RTA-HWD(Fray)
AUTOSAR Module ID	71
AUTOSAR Version	23-11

1.1 Introduction

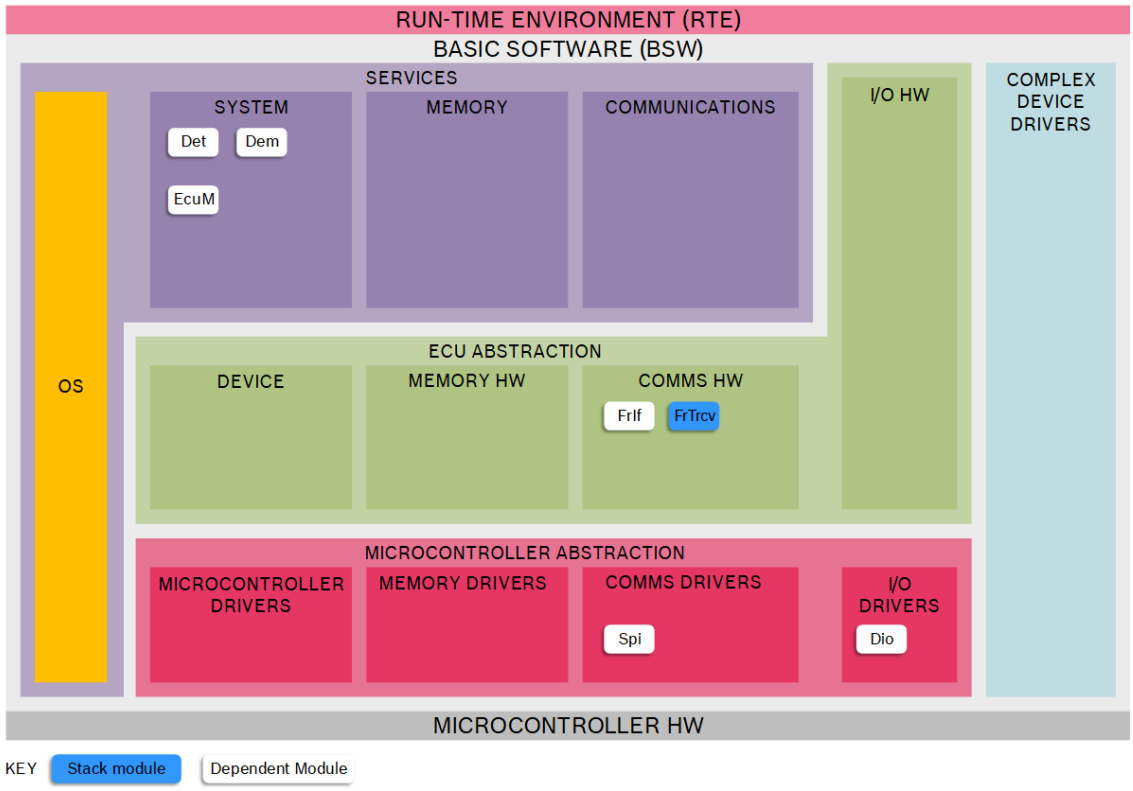

Figure 1.1.FlexRay Architecture

The FlexRay Transceiver Driver module (FrTcrv) is located in the communication hardware abstraction layer, between the communication drivers layer and the upper communication services layer. The diagram above shows the position of FrTrcv in the AUTOSAR layered architecture.

FrTrcv provides a hardware-abstracted, vendor-specific method for controlling FlexRay transceiver devices within an ECU. It is called by the FlexRay Interface (FrIf) utilising a separate FrTrcv instance for each vendor-specific transceiver type. Note that FrIf has no knowledge of which transceiver types exist in a cluster, it is the job of FrTrcv to call the correct driver for each transceiver type.

FrTcrv provides the following functionality:

- Setting of a transceiver's operating mode (defined through logical states as per the AUTOSAR specification).

- Reading the current operating mode (state) for a given transceiver.
- Detection of bus errors encountered when attempting to access a transceiver.
- Transceiver wakeup functions (check, clear, determine reason).
- Enabling/disabling a transceiver branch (for 'active star' network types).
- Returning basic version information for the module.

> **NOTE**
>
> Some functionality is hardware-dependent and may vary according to the vendor-specific transceiver type being accessed.

Module Configuration

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

1.2 Supported Features

1.2.1 Supported Hardware Devices

FrTrcv supports the following hardware devices:

Manufacturer	Device Name
NXP Semiconductors	TJA1080
NXP Semiconductors	TJA1081
NXP Semiconductors	TJA1082
NXP Semiconductors	TJA1085
NXP_Onsemi	TJA1081_NCV7381
Austria Microsystems AG	AS8221
ON Semiconductors	NCV7381
ON Semiconductors	NCV7383
Infineon Technologies	TLE9221

1.2.1.1 Wakeup Support

FrTrcv supports wakeup for all the hardware devices listed above.

Other devices may be supported if they have a specific Complex Device Driver (CDD) interface.

FrTrcv provides API functions to support wakeup if the FrTrcvWakeUpSupport configuration parameter is set to FRTRCV_WAKEUP_BY_POLLING, and if the hardware device is present in the list above.

1.2.1.2 Transceiver Modes

The FrTcrv module is capable of switching a transceiver's operational mode to any of the following logical states:

- POWER_ON
- NOT_ACTIVE
- ACTIVE
- NORMAL
- STANDBY
- SLEEP
- RECEIVE_ONLY

In order to achieve this functionality, the FrTcrv module implements a separate state machine for each specific transceiver device on the FlexRay network. Note that some vendor-specific transceiver types may not support all of the above states. You should consult the vendor's documentation before assuming that all states may be set or detected correctly.

1.3 Extensions

1.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

1.3.1.1 FrTrcv/FrTrcvChannel/FrTrcvAccess/FrTrcvDioAccess/FrTrcv-DioChannelAccess/FrTrcvDioHardwareInterfaceType

Short Name	FrTrcvDioHardwareInterfaceType
Description	Type of functionalty of this pin. The trcv-software uses this parameter to know what functionality is located at the referenced Dio Pin conneted to the Trcv-HW.
Introduction	FRTRCV_HARDWARE_NAME and internal information about ability of this hardware typel.
Multiplicity	1..1
Default Value	-
Enumeration Literals	Enable, Standby, ErrorPin, WakeupSignalisation, Inhibit
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.2 FrTrcv/FrTrcvChannel/FrTrcvAccess/FrTrcvDioAccess/FrTrcv-DioChannelAccess/FrTrcvRbDioSymName

Short Name	FrTrcvRbDioSymName

Description	This parameter have to be configured only if Dio component is not available in the Project. In this case Macro to access Dio will be generated based on the name given here. Macro is generated by attaching the symbolic name pattern. If third party macro is not matching with the generated macro , then a mapping macro should by created by integrator. If project has Autosar Dio then no need to configure this parameter.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.3 FrTrcv/FrTrcvChannel/FrTrcvAccess/FrTrcvDioAccess/FrTrcv-DioChannelAccess/FrTrcvRbHardwarePinInvert

Short Name	FrTrcvRbHardwarePinInvert
Description	Parameter depends on hardware: Describes, if a inverter is between transceiver pin and micro controller. If the transceiver is directly connected to the Controller, 'false' should be configured. If there is an inverter between transceiver and micro controller, 'yes' should be configured.
Introduction	false: No inverter between Micro Controller and transceiver yes: Inverter between Micro Controller and transceiver
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.4 FrTrcv/FrTrcvChannel/FrTrcvAccess/FrTrcvSpiSequence/FrTrcvRb-SpiChnlStrgName

Short Name	FrTrcvRbSpiChnlStrgName
Description	This parameter have to be configured only if SPI component is not available in the Project. In this case Macro to access SPI will be generated based on the name given here.Macro is generated by attachig the symbloic name patter. If generated macro name is not matching with third party macro then integrator should provide the mapping macro .If project has Autosar SPI then no need to configure this parameter.
Multiplicity	0..1

Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.5 FrTrcv/FrTrcvChannel/FrTrcvAccess/FrTrcvSpiSequence/FrTrcvRb-SpiSeqStrgName

Short Name	FrTrcvRbSpiSeqStrgName
Description	This parameter have to be configured only if SPI component is not available in the Project. In this case Macro to access SPI will be generated based on the name given here.Macro is generated by attachig the symbloic name patter. If generated macro name is not matching with third party macro then integrator should provide the mapping macro .If project has Autosar SPI then no need to configure this parameter.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.6 FrTrcv/FrTrcvChannel/FrTrcvAddWaitAfterRead

Short Name	FrTrcvAddWaitAfterRead
Description	Additional wait time for WaitAfterRead
Multiplicity	0..1
Value Range	-100..100
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.7 FrTrcv/FrTrcvChannel/FrTrcvAddWaitBeforeRead

Short Name	FrTrcvAddWaitBeforeRead
Description	Additional wait time for WaitBeforeRead
Multiplicity	0..1
Value Range	-100..100
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.8 FrTrcv/FrTrcvChannel/FrTrcvBranchIdContainer/FrTrcvHardware-BranchId

Short Name	FrTrcvHardwareBranchId

Description	Unique Hardware branch id. Used Brach IDs can be configured here For Example TJA1085 supports upto for Branches 1 2 3 and 4 .
Multiplicity	0..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.9 FrTrcv/FrTrcvChannel/FrTrcvChannelDemEventParameterRefs/ FRTRCV_E_FR_NO_CONTROL_TRCV

Short Name	FRTRCV_E_FR_NO_CONTROL_TRCV
Description	Reference to configured DEM event to report "No/incorrect communication to transceiver&quot
Multiplicity	0..1
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

1.3.1.10 FrTrcv/FrTrcvChannel/FrTrcvChannelDemEventParameterRefs/ FRTRCV_RB_E_UNEXPECTED_MODE_PROP

Short Name	FRTRCV_RB_E_UNEXPECTED_MODE_PROP
Description	Reference to the DemEventParameter which shall be issued when the unexpected mode propogation is detcted in Transceiver HW.
Multiplicity	0..1
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

1.3.1.11 FrTrcv/FrTrcvChannel/FrTrcvChannelDemEventParameterRefs/ FrTrcvRbDynCreate_FRTRCV_E_FR_NO_CONTROL_TRCV

Short Name	FrTrcvRbDynCreate_FRTRCV_E_FR_NO_CONTROL_TRCV
Long Name	Dynamically create DemEvent
Description	Selects the way how the example DemEvent is created. If not set or set to "Disabled", no such DemEvent is automatically created. If the dynamic creation of this DemEvent is enabled, the reference to this DemEvent (in parameter rba_Example_DemEventRef) is also automatically created and shall therefore not be set manually.
Multiplicity	0..1

Default Value	Disabled
Enumeration Literals	Disabled, Standard
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.12 FrTrcv/FrTrcvChannel/FrTrcvEnableInterruptCallout

Short Name	FrTrcvEnableInterruptCallout
Description	This parameter defines the existence and the name of a callout function that enables the interrupt pin for the wakeup. If this parameter is omitted no callout shall take place.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

1.3.1.13 FrTrcv/FrTrcvChannel/FrTrcvHardwareType

Short Name	FrTrcvHardwareType
Description	Type of the used Hardware
Introduction	FRTRCV_HARDWARE_NAME and internal information about ability of this hardware typel.
Multiplicity	1..1
Default Value	-
Enumeration Literals	TJA1080, TJA1081, TJA1082, TJA1085, AS8221, TJA1081_NCV7381, NCV7381, NCV7383, TLE9221, CDD
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.14 FrTrcv/FrTrcvChannel/FrTrcvMasterChannelRef

Short Name	FrTrcvMasterChannelRef
Description	Reference to Master Transceiver where CC Interface is connected.This Parameter is applicable only when TJA1085 transceiver is use. This parameter is also used to Identify wheather CC Interface is connected or Not. If this reference is empty then CC Inter face is enabled and .
Multiplicity	0..1
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

1.3.1.15 FrTrcv/FrTrcvChannel/FrTrcvRbCddChalId

Short Name	FrTrcvRbCddChalId

Description	This Parameter should be configured if Cdd API are having channel ID as one parameter.
Multiplicity	0..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.16 FrTrcv/FrTrcvChannel/FrTrcvRbCddModeChangeDelay

Short Name	FrTrcvRbCddModeChangeDelay
Description	This parameter need to be configured only if unexpected Mode Propogation monitoring for CDD type of Transceiver is enabled. Here user need to specify the minimum delay required to set the requested mode of transcier. User have to enter the number in the multiples of main fucntion duration. For example if Main funtion is scheduled in 10ms task and minimum time required to change the transceiver mode is 20ms then configure this parameter as 2 .
Multiplicity	0..1
Value Range	1..255
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.17 FrTrcv/FrTrcvChannel/FrTrcvRbCddName

Short Name	FrTrcvRbCddName
Description	Only used if CDD is choosen as HardwareType. A separate freetext field which is used to generate function calls out of FrTrcv into the CDD and to include the corresponding .h file. E:G. if “rba_IoExtCy321” is configured, the FrTrcv code generator generates function calls “rba_IoExtCy321_FrTrcvSetOpMode”, “rba_IoExtCy321_FrTrcvGetOpMode”, “rba_IoExtCy321_FrTrcvGetBusWuReason”, “rba_IoExtCy321_FrTrcvSetWakeupMode” and includes the file rba_IoExtCy321.h.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.18 FrTrcv/FrTrcvChannel/FrTrcvTRXDInterfaceUsed

Short Name	FrTrcvTRXDInterfaceUsed
Description	This Tag is applicable only for TJA1085 Transceiver.This tag indicates whether TRXD0/1 interfaces is used to route the signals from one Transceiver to another. In some configuration, If multiple TJA1085 are used where only one instance of TJA1085 is connected to Communication interface and the other one is not connect to Communication interface. The other transciver is used to increase the number of braches by connecting the TRXD/0/1 interfaces as input. User should check the interfacing of tranceiver in schematic and based on the connection this tag need to be set. If this tag is enabled then wakeup on TRXD interface is monitored
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.19 FrTrcv/FrTrcvChannel/FrTrcvTimeoutInner

Short Name	FrTrcvTimeoutInner
Description	Inner Timeout. How long does a read of one bit take until a Interrupt is detected and a new read of statusbit is started. This value is transceiver dependent. If value is 0 or empty, a default value for the specific transceiver is choosen. It is recommended to use default values.
Multiplicity	0..1
Value Range	0..1000
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.20 FrTrcv/FrTrcvChannel/FrTrcvTimeoutOuter

Short Name	FrTrcvTimeoutOuter
Description	Outer Timeout. How long does the SW tryto read out the complete amout of error bit.
Multiplicity	0..1
Value Range	0..10000
Default Value	0

Type	ECUC-INTEGER-PARAM-DEF

1.3.1.21 FrTrcv/FrTrcvGeneral/FrTrcvAutosarVersion

Short Name	FrTrcvAutosarVersion
Description	Naming convention for symbolic names are generated according autosar version: Lecacy: no prefix AR4.0.2: Dio_ (ecuc_sws_2108) AR4.0.3: DioConf_DioChannel_
Multiplicity	1..1
Default Value	-
Enumeration Literals	Legacy, AR4_0_2, AR4_0_3
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.22 FrTrcv/FrTrcvGeneral/FrTrcvClearWupReasonInSetWakeupMode

Short Name	FrTrcvClearWupReasonInSetWakeupMode
Description	On true, the wakeup reason is cleared in function FrTrcv_ClearTransceiverWakeup. With 'false' at this configuration parameter, the wakeup reason is not cleared in function FrTrcv_ClearTransceiverWakeup. The wakeup reason is cleard always init-function and on transition to sleep mode. If this parameter is set to true in one Fr Trcv it will be used in all FrTrcv
Introduction	* true: enabled (ON). * false: disabled (OFF). If switched on, #define FRTRCV_DEV _ERROR_DETECT ON shall be generated. If switched off, #define FRTRCV_DEV_ERROR _DETECT OFF shall be generated. Define shall be part of file FrTrcv_Cfg.h.
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.23 FrTrcv/FrTrcvGeneral/FrTrcvEnableMultipleWakeupreason

Short Name	FrTrcvEnableMultipleWakeupreason

Description	If the enum FrTrcv_TrcvWUReasonType in Fr_Generatypes.h supports multiple wakeupreasons (e.g. FRTRCV_WU_BY_BUS_AND_PIN) then the value should set to true. if the enum FrTrcv_TrcvWUReasonType in Fr_Generatypes.h does not support multiple wakeupreasons (e.g. FRTRCV_WU_BY_BUS_AND_PIN) then the value should set to false (e.g. at Vector stack) If value is set to false, the config parameters FrTrcvRbWakeupReasonBusAndPin, FrTrcvRbWakeupReasonPinAndPower, FrTrcvRbWakeup ReasonBusAndPower, FrTrcvRbWakeupReasonBusAndPinAndPower are relevant.
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.24 FrTrcv/FrTrcvGeneral/FrTrcvGetTrcvWuReasonDirectHwUpdate

Short Name	FrTrcvGetTrcvWuReasonDirectHwUpdate
Description	On true, in function FrTrcv_GetTransceiverWUReason the wakeupreason is directly read out with function FrTrcv_GetTransceiverError. Depending on used transceiver, this can result in gread runtime consumption. On false, in function FrTrcv_GetTransceiverWUReason previous stored wakeupreason is returned. The wakeupreaons is read out at transition to normal mode and at function call FrTrcv_GetTransceiverError. This parameter is global for all configured FrTrcv. If this parameter is set for one transceiver it is used for all FrTrcv.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.25 FrTrcv/FrTrcvGeneral/FrTrcvRbBswmdBooleanSize

Short Name	FrTrcvRbBswmdBooleanSize
Description	At BSWMD generation the bit-size of boolean type has to be forwarded to rba_Arxml_Gen and can be configured here.
Multiplicity	0..1
Default Value	FRTRCV_BSWMDVALUESIZE_8BIT
Enumeration Literals	FRTRCV_BSWMDVALUESIZE_8BIT, FRTRCV_BSWMDVALUESIZE_16BIT, FRTRCV_BSWMDVALUESIZE_32BIT
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.26 FrTrcv/FrTrcvGeneral/FrTrcvRbBswmdEnumSize

Short Name	FrTrcvRbBswmdEnumSize
Description	At BSWMD generation the bit-size of boolean type has to be forwarded to rba_Arxml_Gen and can be configured here.
Multiplicity	0..1
Default Value	FRTRCV_BSWMDVALUESIZE_8BIT
Enumeration Literals	FRTRCV_BSWMDVALUESIZE_8BIT, FRTRCV_BSWMDVALUESIZE_16BIT, FRTRCV_BSWMDVALUESIZE_32BIT
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.27 FrTrcv/FrTrcvGeneral/FrTrcvRbBswmdSpiSize

Short Name	FrTrcvRbBswmdSpiSize
Description	At BSWMD generation the bit-size of boolean type has to be forwarded to rba_Arxml_Gen and can be configured here.
Multiplicity	0..1
Default Value	FRTRCV_BSWMDVALUESIZE_16BIT
Enumeration Literals	FRTRCV_BSWMDVALUESIZE_8BIT, FRTRCV_BSWMDVALUESIZE_16BIT, FRTRCV_BSWMDVALUESIZE_32BIT
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.28 FrTrcv/FrTrcvGeneral/FrTrcvRbCheckWakeupByTrcvReturnVal

Short Name	FrTrcvRbCheckWakeupByTrcvReturnVal
Description	Configures the function declaration of FrTrcv_CheckWakeupByTransceiver False: The function FrTrcv_CheckWakeupByTransceiver behaves as described in Autosar and has the function declaration “void”. True: The function frTrcv_CheckWakeup ByTransceiver has function declaration ‹Std_Return-Type› and returns ‹true›, if wakeup was detected and ‹false›, if no wakeup was detected. No other functionality is changed.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.29 FrTrcv/FrTrcvGeneral/FrTrcvRbWakeupReasonBusAndPin

Short Name	FrTrcvRbWakeupReasonBusAndPin

Description	This parameter is only used if FrTrcvEnableMultiple-Wakeupreason is set to false. If this parameter is used, if wakeupreason by Bus (remote) and by Pin (local) occur at the same time, then the configured wake-upreason is given back in function FrTrcv_ GetBusWuReason This parameter is global for all configured FrTrcv. If this parameter is set for one trans-ceiver it is used for all FrTrcv.
Multiplicity	0..1
Default Value	-
Enumeration Literals	FRTRCV_WU_BY_BUS, FRTRCV_WU_BY_PIN
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.30 FrTrcv/FrTrcvGeneral/FrTrcvRbWakeupReasonBusAndPinAndPow-er

Short Name	FrTrcvRbWakeupReasonBusAndPinAndPower
Description	This parameter is only used if FrTrcvEnableMultiple-Wakeupreason is set to false. If this parameter is used, if wakeupreason by Bus (remote) and by Pin (local) and by Power On occur at the same time, then the config-ured wakeupreason is given back in function FrTrcv_ GetBusWuReason This parameter is global for all configured FrTrcv. If this parameter is set for one trans-ceiver it is used for all FrTrcv.
Multiplicity	0..1
Default Value	-
Enumeration Literals	FRTRCV_WU_BY_BUS, FRTRCV_WU_BY_PIN, FRTRCV_WU_POWER_ON
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.31 FrTrcv/FrTrcvGeneral/FrTrcvRbWakeupReasonBusAndPower

Short Name	FrTrcvRbWakeupReasonBusAndPower
Description	This parameter is only used if FrTrcvEnableMultiple-Wakeupreason is set to false. If this parameter is used, if wakeupreason by Bus (remote) and by Power On occur at the same time, then the configured wake-upreason is given back in function FrTrcv_ GetBusWuReason This parameter is global for all configured FrTrcv. If this parameter is set for one trans-ceiver it is used for all FrTrcv.
Multiplicity	0..1
Default Value	-

Enumeration Literals	FRTRCV_WU_BY_BUS, FRTRCV_WU_POWER_ON
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.32 FrTrcv/FrTrcvGeneral/FrTrcvRbWakeupReasonPinAndPower

Short Name	FrTrcvRbWakeupReasonPinAndPower
Description	This parameter is only used if FrTrcvEnableMultiple-Wakeupreason is set to false. If this parameter is used, if wakeupreason by Pin (local) and by Power On occur at the same time, then the configured wakeupreason is given back in function FrTrcv_GetBusWuReason This parameter is global for all configured FrTrcv. If this parameter is set for one transceiver it is used for all FrTrcv.
Multiplicity	0..1
Default Value	-
Enumeration Literals	FRTRCV_WU_BY_PIN, FRTRCV_WU_POWER_ON
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.33 FrTrcv/FrTrcvGeneral/FrTrcvSPISyncTrnsmitUsed

Short Name	FrTrcvSPISyncTrnsmitUsed
Description	Switches indicates whether SPi Sync Transsmit can be used to Transmit SPI data synchronously.This tag is applicable for TJA1085 transceiver.If SPi does not support sync Trnasmit APi or project does not want to use sync API then set this tag as FALSE
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.34 FrTrcv/FrTrcvGeneral/FrTrcvUseAsr3Api

Short Name	FrTrcvUseAsr3Api
Description	If enabled, the API: FrTrcv_EnableTransceiverWakeup, FrTrcv_Disable-TransceiverWakeup, FrTrcv_Cbk_WakeupByTransceiver is available but the API FrTrcv_CheckWakeupByTransceiver is not available.
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

1.3.2.1 FrTrcv_Cbk_WakeupByTransceiver

Syntax	`FrTrcv_Cbk_WakeupByTransceiver( uint8 FrTrcv_TrcvIdx )`
Parameters In	`FrTrcv_TrcvIdx` Identifies the Transceiver.
Description	Informs upper layers on a wakeup event. If the current Trcv wakeup state/pin has to be checked, FrTrcv_ClearTransceiverWakeup has to be called first to clear old wakeups.

1.3.2.2 FrTrcv_CheckModePropagation_Dio

Syntax	`FrTrcv_CheckModePropagation_Dio(uint8 FrTrcv_TrcvIdx)`
Parameters In	`FrTrcv_TrcvIdx` Identifies the Transceiver.
Description	Checks latest mode with the curent mmode of the Transceiver, and if the latest requested mode does not match the current mode, the function reports a DEM Error.

1.3.2.3 FrTrcv_ChangeModeFunction3

Syntax	`FrTrcv_ChangeModeFunction3(uint8 FrTrcv_TrcvIdx)`
Parameters In	`FrTrcv_TrcvIdx` Identifies the Transceiver.
Description	Reads wakeup reason by evaluation of FrTrcv_GetTransceiverError and sets FrTrcv_Dio_WakeupReason_au8. This function is called before TJA1080, TJA1081, TJA1082, NCV7383 reaches normal mode.

1.3.2.4 FrTrcv_CheckWakeupInternal_CDD

Syntax	`FrTrcv_CheckWakeupInternal_CDD(uint8 FrTrcv_TrcvIdx, uint8 CddNo)`
Parameters In	`FrTrcv_TrcvIdx` Identifies the Transceiver. `CddNo` Cdd Transciever number to acess the Dio.
Parameters Out	`RxStats` list of statistic counters defined with Eth_RxStatsType for corresponding port of the switch.
Description	Reads out configured Dio Pin for Cdd type of transceiver. If the Pin level is Low, then it is treated as a valid wakeup, otherwise no wakeup is detected. This function is available only when Dio is configured for a CDD type Transceiver.

1.4 Limitations

The following features are not supported for version v12.7.0:

There are no unsupported features in this release of the module.

1.5 Exclusions

There are no unsupported API functions in this release of the module.

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- FrTrcv/FrTrcvChannel/FrTrcvChannelEcucPartitionRef
- FrTrcv/FrTrcvChannel/FrTrcvChannelUsed
- FrTrcv/FrTrcvChannel/FrTrcvControlsPowerSupply
- FrTrcv/FrTrcvChannel/FrTrcvIcuChannelRef
- FrTrcv/FrTrcvChannel/FrTrcvMaxBaudrate
- FrTrcv/FrTrcvGeneral/FrTrcvDemReportErrorStatusConfiguration
- FrTrcv/FrTrcvGeneral/FrTrcvEcucPartitionRef
- FrTrcv/FrTrcvGeneral/FrTrcvTimerType
- FrTrcv/FrTrcvGeneral/FrTrcvWaitTime

1.6 Deviations

1.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

1.6.1.1 FrTrcv/FrTrcvChannel/FrTrcvAccess

[ADDED] Description	FrTrcvAccess

1.6.1.2 FrTrcv/FrTrcvChannel/FrTrcvBranchIdContainer/FrTrcvBranchId

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

1.7 API Definition

List of supported API functions

Table 1.43.FrTrcv Supported API functions

`FrTrcv_Init()`	Initialises the module
`FrTrcv_SetTransceiverMode()`	Sets the operating mode for the specified transceiver

`FrTrcv_GetTransceiverMode()`	Returns the current operating mode of the specified transceiver
`FrTrcv_GetTransceiverWUReason()`	Returns the wakeup reason information for the specified transceiver
`FrTrcv_GetVersionInfo()`	Returns version information for the module
`FrTrcv_GetTransceiverError()`	Returns detailed information on the error state of a specified transceiver
`FrTrcv_DisableTransceiverBranch()`	Disables the specified transceiver branch on an 'active star' network
`FrTrcv_EnableTransceiverBranch()`	Enables the specified transceiver branch on an 'active star' network
`FrTrcv_ClearTransceiverWakeup()`	Clears a pending wakeup event
`FrTrcv_CheckWakeupByTransceiver()`	Triggers a notification to upper layer software that a 'bus wakeup' event has been detected

For Further information please refer to AUTOSAR_SWS_FlexRayTransceiverDriver.pdf

1.8 Configuration Advice

NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

1.8.1 Transceiver Configuration

Transceivers are identified by a unique ID, which is set on a per-channel basis using the configuration parameters contained in `FrTrcvChannel`. Therefore, you should configure multi-channel transceivers with a separate set of parameters for each channel (see Multi-Channel Transceivers below).

For transceivers which use the DIO interface, configure the DIO port for the port pin which is connected to the transceiver's Enable, Standby or Error pin. This information will be available in the vendor's ECU schematic. These settings are made under the `FrTrcvDioAccess` parameters.

1.8.1.1 Multi-Channel Transceivers

Some transceivers have multiple channels, and the name of these transceivers may not be found directly in the transceiver HW list. You should refer to the transceiver's user manual to identify the names of individual transceiver channels, which will need to be provided during configuration.

For example, the TJA1901 is actually: TJA1046 + TJA1029 (2 x TJA1044GT +

TJA1029).

TJA1901 will not be available as a name in the transceiver list, but TJA1044 and TJA1029 will be available, and these should be selected instead of TJA1901.

1.8.2 CDD Channel Id

For CDD Transceivers, the CDD Channel Id must either be configured for all the available CDD Trcvs, or it must not be configured for any of them.

1.8.3 Wakeup By Polling

In order to configure the wakeup by polling feature, set parameter `FrTrcvWakeUpSupport` to the value FRTRCV_WAKEUP_BY_POLLING. You should check that the transceiver hardware type supports this wakeup type before enabling this feature.

1.8.4 Initialisation State

To configure the state of a transceiver after initialisation (through the API function `FrTrcv_Init`), set the configuration parameter `FrTrcvInitState` to one of the following states:

- FRTRCV_TRCVMODE_NORMAL
- FRTRCV_TRCVMODE_RECEIVEONLY
- FRTRCV_TRCVMODE_STANDBY
- FRTRCV_TRCVMODE_SLEEP

If not explicitly set, this parameter defaults to a value of FRTRCV_TRCVMODE_NORMAL.

1.8.5 TJA1085 transceiver

When configuring TJA1085 Trcv hardware, update the MCAL pin information to be connected to the RSTN pin and also update the time required for resetting.

1.9 Integration Advice

1.9.1 Init Functions

FrTrcv_Init initializes FrTrcv, and the transceiver mode is set as per the FrTrcvInitState configuration parameter.

1.9.2 DeInit Functions

No DeInit is required.

1.9.3 Scheduled Functions

FrTrcv_MainFunction is called at a cyclic interval defined by the configuration parameter FrTrcvMainFunctionCycleTime. If a cycle time of 0 is configured, FrTrcv_Mainfunction is not executed by the scheduler and need not be present in compiled code.

1.9.4 Wakeup Reasons

To support multiple wakeup reasons, the function `FrTrcv_GetTransceiverWUReason()` returns a bit pattern in which multiple bits may be set. Therefore, when checking for wakeup reasons using this function, your code must use bitwise (rather than byte-level) operators to correctly interpret the reason data.

1.9.5 SPI Communication

When using `FrTrcv_GetTransceiverMode()` to read the current state of a transceiver through an SPI communication driver (e.g. for TJA1085 transceiver type), you may need to call the function twice in order to obtain the state information. This is due to the asynchronous nature of SPI communication.

If your project uses SPI communication drivers, you should use the FrTrcv_Cfg_IntCode.h header file to map to the correct API functions of the SPI module and to define the timeout parameters for SPI communication.

1.9.6 Active Star Network Using TJA1085

If your implementation includes an 'active star' network comprising multiple TJA1085-type transceivers, the following considerations apply:

- All the transceivers are set to the mode requested for the transceiver where the CC interface is connected ('primary' or 'master' transceiver).
- `FrTrcv_GetTransceiverMode()` returns the mode of the 'primary' or 'master' transceiver irrespective of the mode of the 'secondary' or 'slave' transceiver.
- Wakeup detection is based on the consolidated data of the 'primary' or 'master' transceiver AND the 'secondary' or 'slave' transceiver.
- To control branches, read error information, or return the wakeup reason, the 'primary' or 'master' transceiver should be addressed separately from the 'secondary' or 'slave' transceiver.

Note: TJA1080 can not be used as star.

1.9.7 Dependency on rba_Trcv

FrTrcv's dependency on the rba_Trcv module has now been removed, and so rba_Trcv is not also required in order to generate the code for the FrTrcv module.

Now that FrTrcv works independently of rba_Trcv, if you were using any rba_Trcv APIs, these APIs must be replaced with the equivalent APIs of FrTrcv.

The template file FrTrcv_Cfg_Time.h has to be adapted (hints for adapting FrTrcv_Cfg_Time.h are in the file's comments).

1.9.8 Memory Mapping

If your project requires memory mapping support, you should use the FrTrcv_MepMap.h header file to map to the correct BSW memory mapping macros. If your project doesn't require memory mapping support, then the macros defined

in FrTrcv_MemMap.h should be mapped to empty values instead.

1.9.9 Additional Update Hints

The API FrTrcv_PreInit_TJA1085 is used to read the wakeup information before the initialization.

1.9.10 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The FrTrcv module uses the following OS resources:

- FrTrcv_RESOURCE

1.10 Hardware Requirements

Table 1.44.FrTrcv Hardware Requirements

Support	Usage
FLASH	3052 bytes
RAM	39 bytes

Usage calculated based on a sample configuration.

2 Glossary

Active Star
A network topology consisting of one or more active central nodes which rebroadcast transmissions received from a branch to all other branches on the network

Branch
An element of an 'active star' network electrically connected to the same transmitter and receiver circuit on the physical layer

CC FlexRay Communication Controller

CDD
Complex Device Driver

Cluster
A communication system made up of a number of nodes linked by a physical transmission medium

FrIf AUTOSAR FlexRay Interface module

FrNm
AUTOSAR FlexRay Network Management module

FrSM
AUTOSAR FlexRay State Manager module

FrTp
AUTOSAR FlexRay Transport Layer module

L-PDU
PDU of the data link layer

Node
An ECU which is connected to a FlexRay bus via the FlexRay Interface

PDU
Protocol Data Unit